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FEDERAL SPECIFICATION

BOILERS, STEAM WATERTUBE (BENT TUBE, MULTI-DRUM AND CROSS DRUM) PACKAGED TYPE (10,000,000 TO 125,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers watertube (bent tube, multi-drum, and cross drum) packaged boiler units.

The boilers covered by this specification are intended for generation of steam to be used in processing operations and space heating systems in land installations.

1.2 Classification. The boilers are of the following types, sizes, and classes (see 6.2).

TYPES

Type I - Steam boiler producing low pressure steam not greater than 15 pounds per square inch gage pressure (psig) (103 kilopascals gage (kPag)).
Type II - Steam boiler producing high pressure steam between 16 and 150 psig (111 and 1034 kPag).
Type III - Steam boiler producing high pressure steam between 151 and 350 psig (1 041 and 2413 kPag).
Type IV - Steam boiler producing high pressure steam between 351 and 600 psig (2420 and 4137 kPag).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 1581), Naval Construction Battalion Center, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SIZES

Size 3 - Boiler steam output capacity between 10 000 and 36 100 pounds per hour (PPH) (4536 and 16 375 kilograms (troy) per hour (kgph)).
Size 4 - Boiler steam output capacity between 36 101 and 125 000 PPH (16 375 and 56 699 kgph).

CLASSES

Class 1 - Heavy oil fired.
Class 2 - Light oil fired.
Class 3 - Gas fired.
Class 4 - Combination heavy oil and gas fired.
Class 5 - Combination light oil and gas fired.
Class 6 - Coal fired.
Class 7 - Combination coal and oil fired.
Class 8 - Combination coal and gas fired.
Class 9 - Combination coal, oil, and gas fired.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

F-B-2910 - Burners, Single: Oil, Gas, and Gas-Oil Combination, for Packaged Boilers (320,000 to 125,000,000 Btu/hr Thermal Output).
F-S-2911 - Stokers, Mechanical: for Packaged Boilers Fired on Coal or a combination of Coal-Gas, Coal-Oil, or Coal-Gas-Oil (2,000,000 to 50,000,000 Btu/hr Thermal Output Capacity).
OO-B-2925 - Baghouse, Reverse Pulse Jet; For Packaged Boilers Fired On Coal or Combination of Coal and Gas, Coal and Oil, or Coal-Gas-Oil (2,000,000 to 125,000,000 Btu/hr Thermal Output Capacity).

Military Standard

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Equipment.

(Copies of military and federal specifications required by contractors in connection with specific procurement functions are obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.

Section I	-	Power Boilers, Rules for Construction of.
Section IV	-	Heating Boilers, Rules for Construction of.
Section VIII Div. 1	-	Pressure Vessels, Rules for Construction of.
Section IX	-	Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

ASME Performance Test Code, PTC 4.1 - Steam Generating Units.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM E 427	-	Testing for Leaks Using the Halogen Leak Detector.
ASTM E 515	-	Test Method for Leaks Using Bubble Emission Techniques.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1	-	General Standards for Industrial Control and Systems.
NEMA ICS 2	-	Standards for Industrial Control Devices, Controllers, and Assemblies.
NEMA ICS 6	-	Enclosures for Industrial Control and Systems.
NEMA MG 1	-	Motors and Generators.

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	-	National Electrical Code.
NFPA 8502	-	Prevention of Furnace Explosions/Implosions in Multiple Burner Boiler-Furnaces.

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(Private sector and civil agencies may purchase copies of these voluntary standards from the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL-353 - Limit Controls.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA, 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The boiler shall be a packaged watertube style with a bent tube, multi-drum, or cross drum design. The boiler shall be capable of producing the steam output as specified (see 6.1 and 6.2). The boiler shall be furnished complete with all appurtenances specified herein and shall be factory-wired and ready for operation when connected to water, fuel, and electrical supplies. The boiler shall be fired on the fuel specified (see 6.1 and 6.2). The boiler shall be furnished complete with the burner specified in F-B-2910, or the stoker specified in F-S-2911, or both as specified (see 6.2).

3.2 Codes and standards. The boiler shall conform to the applicable requirements of Section I; Section IV; Section VIII, Division 1; and Section IX of the ASME Boiler and Pressure Vessel Code (hereinafter referred to as the ASME Code), NFPA 70, NFPA 85C. Limit controls and interlock switches shall be in accordance with UL-353. Electrical wiring design practices shall be installed in accordance with the applicable requirements of NFPA 70. Electric motors, starters, and controls shall conform to the applicable requirements of NEMA ICS 1, NEMA ICS 2, NEMA ICS 6, and NEMA MG-1.

3.3 Standard commercial product. The boiler shall, as a minimum, be in accordance with the requirements of this specification and may be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, may be included in the boiler being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements of manufacturer's catalogs, or brochures, and represents the latest production model.

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3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The boiler shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation. The boiler shall be designed for the following operating parameters as specified (see 6.2):

a. Design pressure (psig (kpag)).
b. Operating pressure of steam at packaged unit outlet (psig (kpag)).
c. Design temperature (degrees Fahrenheit (°F) (degrees Celsius (°C)).
d. Operating temperature of steam at packaged unit outlet (°F (°C)).
e. Maximum continuous steam output (PPH (kgph)).
f. Feedwater temperature (°F (°C)).
g. Maximum, minimum, and average anticipated ambient air temperatures (°F (°C)).
h. Site elevation above sea level (feet (millimeters (mm)).

3.6.1 Stoker retrofit design. When specified (see 6.1 and 6.2), the design of the boiler shall be such as to allow the boiler to be retrofitted with a stoker if a stoker is not currently required. Boilers designed to be retrofitted shall have maximum flue gas velocities and furnace exit temperatures in the range typical of other coal fired boilers. Tubes in the convection section shall be designed to be in line. If the maximum steam output capacity of the stoker fired boiler is less than the steam output capacity of 3.6, then the instructions shall also include how supplemental oil or gas burners shall be used to attain the required steam output capacity of 3.6. Supplemental oil or gas shall be limited to 40 percent of the required steam output capacity.

3.7 Performance characteristics.

3.7.1 Steam purity. The moisture content of the steam leaving the steam drum shall be not greater than 0.5 percent for the boiler water total dissolved solids (TDS) concentrations and the steam pressure indicated in table I, when operated at 100 percent load, maximum continuous rating.

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3.7.2 Fuel-to-steam efficiency. When tested in accordance with 4.7.2, the fuel-to-steam efficiency shall be not less than 81 percent for light and heavy oil fuels, 77 percent for gas fuel, and 80 percent for coal fuel. These efficiency requirements are independent of an auxiliary heat recovery unit and are the lowest they can be at any firing rate.

3.7.3 Furnace heat input rate. When the boiler is operating at maximum output, the heat input rates to the furnace shall be not greater than the values specified in 3.7.3.1 through 3.7.3.3.

TABLE I. Boiler water quality and associated steam purity.

Steam Drum Pressure (psig)		Boiler Water TDS (parts per million (ppm), maximum)
0 - 300	(0 - 2068 kPag)	3500
301 - 450	(2075 - 3103 kPag)	3000
451 - 600	(3110 - 4137 kPag)	2500

Note: Boilers with superheaters may require one ppm carryover.

3.7.3.1 Gas and oil fired equipment volume heat input rate. The volume heat input rate for gas and oil fired boilers shall be not greater than 70 000 British thermal units per hour per cubic feet (Btu/hr/ft^3) of furnace volume.

3.7.3.2 Coal fired equipment volume heat input rate. Unless otherwise specified (see 6.2), the volume heat input rate for coal fired boilers shall be not greater than 30 000 (Btu/hr/ft^3) except for boilers firing lignite or sub-bituminous coal. Boilers firing lignite or sub-bituminous coal shall have a volume heat input rate not greater than 45 000 (Btu/hr/ft^3) (see 6.1.1).

3.7.3.3 Surface heat input rate. The maximum heat input rate in Btu/hr per square foot (ft^2) ((Watts) per square meter (m^2)) of effective radiant heating surface of the furnace shall be not greater than the values of table II.

TABLE II. Maximum allowable heat input rate.

Rated Capacity, thousand Btu/hr		Btu/hr/ft^2
less than 36 100 (10 573 Watts)		80 000
36 101 to 63 000	(10 573 - 18 451 Watts)	100 000
63 001 to 95 000	(18 451 - 27 823 Watts)	115 000
95 001 to 125 900	(27 283 - 36 872 Watts)	125 000

3.7.3.4 Stoker fired boiler grate heat release rate. The maximum heat release rate per ft^2 (m^2) of effective air admitting grate area shall be not greater than the values given in table III.

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TABLE III. Maximum allowable stoker grate heat input rates.

Stoker Design	Maximum Grate Heat Release Rate Btu/hr/ft^2
Spreader stoker	700 000
Overfeed stoker	450 000
Underfeed stoker	450 000

3.7.4 Exit temperature at stack. The exit temperature of the flue gas at the stack shall be not less than 325 °F (163 °C) without an economizer. When exit temperatures of less than 325 °F (163 °C) without an economizer are allowed, the flue gas passages shall be of corrosion-resistant materials to prevent potential flue gas condensate corrosion problems. When operating at minimum load and an economizer is required (see 3.17 and 6.2), the exiting flue gas temperature shall be not less than 300 °F (149 °C). The economizer water temperature at flue gas exit shall be not less than 220 °F (104 °C). When exit flue gas temperatures of less than 300 °F (149 °C) are allowed, use corrosion-resistant materials in the flue gas passages and economizer to prevent potential flue gas condensate corrosion problems.

3.7.5 Baghouse design requirements. When specified (see 6.2), the boiler manufacturer shall provide the following information to be used for the design of the flue gas baghouse in accordance with OO-B-2925:

a. Maximum gas flow in standard ft^3/min (mm^3/min).
b Required draft at boiler flue gas outlet in inches (mm) of water at maximum gas flow conditions.
c. Dimension of flue gas outlet.
d. Height of flue gas outlet above grade and relative orientation.
e. Flue gas outlet particulate loading.
f. Flue gas outlet temperature ranges (including effects of all exchangers such as economizers, etc.).

3.7.6 Thermal shock. The steam boiler or hot water generator shall be designed to withstand the thermal stresses associated with cold start and emergency shutdown operating conditions.

3.8 Environmental requirements. The emission requirements shall be met at the maximum required continuous output. The boiler and burner shall meet the legal emission requirements required by the Federal, state, and local environmental rules and regulations. The emission requirements for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide shall be as specified (see 6.1 and 6.2).

3.9 Construction. Boilers shall be constructed for the design conditions and performance requirements specified herein and in accordance with the applicable sections of the ASME Code.

3.9.1 Drums. Steam drums shall provide stable water drum level, shall provide adequate water circulation, and shall be of adequate size to allow maintenance access to the baffles and other

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internals. Steam baffles shall be removable. A chemical injection header shall be provided. No parts of a boiler drum or header surface shall be directly exposed to furnace radiation unless it is adequately protected by a refractory of a water screen in such a manner that no part of the exposed drum or header surface shall be more than 2 inches (50 mm) from the nearest water tube.

3.9.2 Tubes. Tubes shall be either electric-resistance welded or seamless steel. Tube bends shall be performed in a manner to permit cleaning of the internal surface of tubes with conventional tube-cleaning tools. Unless otherwise specified (see 6.2), radiant tube clearances shall be not greater than 1-inch (25 mm) with membrane or finned tube construction. Spinners, turbulators, and other such devises shall not be permitted in the tubes.

3.9.3 Refractories. Refractories shall withstand the temperature existing under maximum load conditions. The refractories shall be formed in place or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment.

3.9.4 Access door and observation ports. The boiler shall be equipped with a refractory-lined access door for internal inspection and repairing of the furnace fireside. The boiler shall be equipped with cover plates or handholes to allow for internal inspection and cleaning. Not less than two heat resistant observation ports shall be furnished to allow observation of combustion conditions. Observation ports shall be fitted either with safety shields to maintain closure unless held open by the operator, or with air-cooled heat-resistant glass closures. Access to the furnace may be through the burner opening provided it has an unrestricted opening of not less than 15 inches (381 mm) in diameter. Access door and observation ports shall be gas-tight.

3.9.5 Baffles. Baffles, as required, shall be arranged to diffuse the gas properly over heating surfaces in order to obtain maximum heat absorption. The baffles shall be held securely in place without being affected by or interfering with the free expansion or contraction of the boiler. If the baffles are steel, they shall be not less than 0.25-inch (6 mm) thick, or heat resistant alloy they shall be not less than 0.125-inch (3 mm) thick. Tangent tube baffles may be used in lieu of metal plate baffles. Baffles shall withstand the temperature under maximum load conditions.

3.9.6 Furnace. The walls and floor of the furnace shall permit free expansion of the boiler without undue stress on any part of the boiler. The furnace side walls and roof shall be water-cooled type construction. Water wall headers and supply tubes shall be protected from the products of combustion. Unless otherwise specified (see 6.2), the furnace shall be not greater than 35 feet (10.67 m) in length.

3.9.7 Insulation. All exterior surfaces of the boiler, including the floor, but excluding doors, drumheads, handholes, manholes, vents, and appurtenances, shall be insulated and jacketed. No asbestos material shall be used. The insulation shall be so formed and secured in place as to prevent sagging or displacement during shipment and operation. The insulation shall be covered with a sheet metal jacket made of steel having a thickness not less than 0.023-inch (0.58 mm) (24 Standard Gage) or nonferrous sheet metal having a thickness not less than 0.029-inch

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(0.74 mm) (22 Standard Gage). All sections of the boiler exposed to the products of combustion shall be backed by refractory, by insulation, or both, to maintain a casing temperature of not greater than 130 °F (54 °C) with a surface air velocity of 5 feet (1.52 m) per second and an ambient temperature of 80 °F (27 °C) while the boiler is operating at maximum capacity.

3.9.8 Inner casing. When an inner casing is part of the manufacturer's standard design, the inner casing shall be pressure-tight to prevent the products of combustion from coming in contact with the boiler insulation and outer casing. The inner casing may either be the welded wall itself or a separate inner casing having a thickness not less than 0.135-inch (3.43 mm), placed just behind the tubes or the refractory backing. The casing shall be pressurized for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches (508 mm) of water, whichever is less. The pressure drop shall be not greater than 5 inches (127 mm) of water in ten minutes.

3.9.9 Base. The boiler shall be mounted on steel supports fastened securely to a structural steel base. The base shall be capable of supporting the weight of the unit and its contained water.

3.10 Boiler trim. Each boiler shall be fully equipped with the trim required under the applicable section of the ASME Code for the type of boiler specified herein. The additional appurtenances specified herein shall also be furnished. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The trim, valves, connecting piping, and fittings shall be suitable for the design pressure and temperature as specified herein. Limit controls and interlock switches shall be in accordance with the applicable requirements of NFPA 8502, and UL-353. Signals shall be compatible with the control system required in F-B-2910 or F-S-2911. Unless otherwise specified (see 6.2), the trim and appurtenances shall include but not necessarily be limited to the following:

a. High water level alarm.
b. Low water level alarm and a low water level fuel cutoff device. The device shall be mounted on separate connections and be provided with a blowdown valve.
c. A manual bypass momentary contact push button switch for the low water level cutoff switch of item b shall be located adjacent to the drain valves for the low water cutoff switch.
d. Piping and valves for continuous blowdown. The continuous blowdown valve shall have stainless steel trim and be able to maintain the position of valve setting.
e. A steam drum vent valve and corrosion-resistant trim.
f. A chemical feed pipe connection with a diameter of not less than 0.75-inch (19 mm), nominal with a corrosion-resistant trimmed stop valve installed on the top drum and on the feedwater inlet piping.

3.11 Feedwater regulator. Steam boilers shall be furnished with an automatic feedwater regulator system to maintain the boiler manufacturer's established boiler steaming water level. The regulating system shall maintain the setpoint within ±5 percent of the full range of allowable operating level. Feedwater regulators shall include the required sensing, actuating, regulating, and controlling equipment necessary to perform under the required conditions specified herein. The

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regulating system shall be provided complete with all necessary fittings, tubing, piping, mechanical linkages, wiring, and other material or parts required to connect the operating components.

3.11.1 Feedwater regulating valve. Regulating valves shall be furnished with a double block and bleed piping bypass to allow the regulator valve to be removed while the boiler is operating. The regulating valves shall be equipped with a handwheel or knob for manual control and operation and shall have the manufacturer's standard position indicator. The valve shall be constructed with a balanced force design to allow the valve to move easily against the force of the feedwater flow. Materials of construction shall be the manufacturer's standard for good corrosion resistance in boiler feedwater service.

3.11.2 Manual feedwater regulator control. When specified (see 6.2), the feedwater regulator system shall be furnished with a manual and automatic switch and a controlled mechanism for manually adjusting the feedwater regulator position. The direction in which the control mechanism should be operated to raise or lower the water level shall be clearly indicated.

3.12 Burners and stokers. Class l, class 2, class 3, class 4, class 5, class 7, class 8, and class 9 boilers shall be furnished with factory installed burners, ignition systems, and associated fuel systems in accordance with F-B-2910. Class 6, class 7, class 8, and class 9 boilers shall be provided with a stoker in accordance with F-S-2911.

3.13 Combination fuel boilers. Class 7, class 8, or class 9 boilers are to be designed in accordance with all requirements of F-S-2911 for the mechanical stoker and all requirements of F-B-2910 for the oil, gas, or oil and gas burner. In case of conflict or duplication of equipment specified, F-B-2910 shall take precedence. The system shall be integrated to operate simultaneously. A common control panel for both the burner and the stoker shall be provided.

3.14 Draft fans and primary air supply systems. The draft fan for burners shall conform to F-B-2910. Primary air supply systems for stokers shall conform to F-S-2911 as applicable.

3.15 Combustion control system. The boiler shall be furnished with combustion control systems conforming to F-B-2910 and F-S-2911 as applicable. The boiler firing rate shall be controlled as a function of the boiler outlet steam pressure.

3.15.1 Burner combustion control system. The burner combustion control system shall be in accordance with F-B-2910. When specified (see 6.1 and 6.2), the combustion control system shall be furnished with an oxygen compensation system, or when specified (see 6.2), an oxygen compensation system with an unburned combustible gas analyzer. The combustion control system shall be one of the following, as specified (see 6.2):

a. Single point positioning control.
b. Parallel positioning control.
c. Metering control.

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3.15.2 Stoker combustion control system. The stoker combustion control system and control linkages shall be in accordance with F-S-2911. The boiler manufacturer shall furnish automatic combustion controls for the stoker control linkages. The controllers shall maintain the fuel and air input in accordance with thermal load demand. The relative rates of change shall be controlled simultaneously so as to maintain the air to fuel ratio within the limits required for continuous combustion and stable furnace pressure throughout the controllable operating range of the stoker.

3.15.3 Combustion control system for combination fuel boilers. The combustion control system for class 7, class 8, and class 9 boilers shall be in accordance with 3.15.1 and 3.15.2. The stoker combustion system shall maintain a constant fuel and air input at or below a variable manual set limit. The gas or oil burner thermal input shall be modulated as necessary to supply the additional thermal input required to meet the thermal load demand.

3.16 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and control system circuits shall be mounted on a single control panel or cabinet insofar as practicable in order to centralize the control functions. The location of the panel shall either be at the side of the boiler, or in a free-standing control panel for remote mounting as specified (see 6.2). When a free-standing remote mounted control panel is to be furnished, skid wiring or tubing shall terminate in a skid mounted junction box.

3.17 Economizer. When specified (see 6.2), a separately packaged economizer shall be furnished. The economizer shall be complete with prefabricated interconnecting breaching for attachment to the boiler flue gas outlet and to the feed piping between the economizer and the boiler drum and shall require minimum field assembly. The economizer shall be equipped to heat feedwater. An automatic feedwater preheat system shall be provided when necessary to prevent potential corrosion or flue gas condensation. Temperature gages shall be furnished on the inlet and outlet piping of the flue gas and the feedwater. Manual valving shall also be provided to allow the feedwater to bypass around the economizer. The valving and piping shall be designed so that there is no shutoff valve between the economizer outlet and the steam drum inlet to eliminate the need for a separate economizer relief valve. Economizers specified with class 1, class 4, class 6, class 7, class 8, and class 9 boilers shall be furnished with soot blowers in conformance with 3.18. Provisions shall be made for future installation of soot blowers on economizers specified with class 2, class 3, or class 5. The economizer shall be designed and constructed in accordance with the applicable section of the ASME Code. Materials used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions.

3.18 Soot blowers. Soot blower(s) shall be furnished with class 1, class 4, class 6, class 7, class 8, or class 9 boilers. Provisions shall be made for future installation of soot blowers on class 2, class 3, and class 5 boilers. When specified (see 6.2), automatically controlled sequencing soot blowers shall be provided. Soot blowers shall be steam operated, valve-in-head type and shall be furnished complete with wall sleeves, clamps, hangers, supports, operating chains, and other appurtenances required for a complete installation. The blower elements shall be so arranged that the heating surfaces shall be cleaned of any soot deposits when rotated by manual means. Elements shall be of such length, diameters, and total nozzle area that for the operating pressure

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involved there will be no significant difference in the cleaning effect between the nozzle nearest the inlet and those farthest from the inlet of the element. Soot blower control valve shall include a valve chain operated from the floor, in addition to the auxiliary steam-stop valve and drain connection. Soot blowers shall be fitted with scavenging air ports and shall be connected by tubing to a draft source in a manner to provide sufficient air for continuous scavenging of the blowers while the boiler is firing during periods when soot blowers are not operated for cleaning purposes. Soot blowers shall be deactivated upon a boiler shutdown and require a manual reset.

3.19 Fuel oil preheaters. When specified (see 6.1 and 6.2), fuel oil preheating systems shall be furnished as part of the packaged boiler in addition to the trim preheater described in F-B-2910. The heater shall be either electric or steam as specified (see 6.2). The heater and associated controls may be supplied either mounted on the boiler assembly or supplied separately for installation by the mechanical contractor. The preheating equipment shall be designed in accordance with Section VIII, Division 1 of the ASME Code. The system shall be designed to heat 120 percent of the maximum fuel oil flow from the specified oil supply temperature (see 6.2) to the required atomization temperature. Pressure drop across the preheater shall be not greater than 15 psi (103 kpag). When the required fuel oil temperature is reached and maintained by the heater, a thermostatic control or limit switch shall cutoff the heater. A high temperature preheater cutoff switch shall also be provided with a manual reset. Temperature gages shall be provided to measure fuel oil inlet and outlet temperatures across the preheater.

3.19.1 Electric preheater. When the electric preheater is required (see 3.19), the preheater shall be furnished with a temperature controller to maintain the fuel oil temperature within 5 percent of the setpoint temperature. Preheater operating voltage shall be not greater than 230 volts (V). Electric preheaters rated up to 1800 watts (W) (6 146 Btu/hr) shall be controlled by a line voltage thermostat switch or by a magnetic contactor with a coil operated by a thermostat. Preheaters rated at greater than 1800W (6 146 Btu/hr) shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. The electric power and separate control circuits shall be protected by a circuit breaker located on the boiler assembly.

3.19.2 Steam preheater. When the steam preheater is required (see 3.19), the heater shall be designed for the boiler design pressure specified in 3.6 and shall use steam generated by the boiler. The preheater shall be furnished with an adjustable oil temperature regulator to maintain the fuel oil temperature within 5 percent of the setpoint temperature. Means shall be provided for visually detecting oil in the preheater condensate line.

3.19.2.1 Electric preheater within steam preheater. When specified (see 6.2), the steam preheater shall be furnished with an electric preheater in the same exchanger shell for cold start up. The electric preheater shall have sufficient capacity for the firing rate occurring when the combustion controls are in the position for low fire starts. The electric preheater shall be furnished in accordance with the requirements of 3.19.1.

3.20 Cinder return system. When a class 6, class 7, class 8, or class 9 boiler is required, a cinder return system shall be furnished if specified (see 6.2). The cinder return system shall include all necessary conveyor lines, nozzles, ducts, dampers, and other equipment for returning the cinders

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from settling chamber hoppers, under the boiler drums, and other internal locations where cinder buildup will occur. All cinder conveyor lines, elbows, and cinder pickup nozzles shall have a nominal inside diameter of not less than 3 inches (75 mm). Construction material shall be abrasion resistant.

3.21 Transition piece. When specified (see 6.2) , the manufacturer's standard transition piece shall be provided.

3.22 Electrical requirements. Electrical wiring design practices shall be installed in accordance with the applicable requirements of NFPA 70. The boiler and associated equipment shall obtain power from a circuit with a single master manual service disconnect switch and automatic circuit breaker for all boiler electrical equipment. Motors shall be in accordance with the provisions of NEMA MG 1, shall be totally enclosed fan cooled, and designed for operation on the voltage, phase, and frequency as specified (see 6.2). Unless otherwise specified (see 6.2), motors of 100 horsepower (74 600W) and less shall be provided with magnetic across-the-line starter and overload protection. Motors over 100 horsepower (74 600W) shall have reduced voltage starting. Motor starters shall conform to the applicable requirements of NEMA ICS 1 and ICS 2. Motor starters and controls shall he enclosed in NEMA ICS 6, type 12 enclosures and may be installed in the control panel of 3.16.

3.23 Cleaning, treatment, and painting. Bare steel surfaces shall be cleaned and a primer coat applied. Paint shall be applied to normally painted surfaces in good commercial practice.

3.24 Lifting and tiedown attachments. When specified (see 6.2), the boiler shall be equipped with lifting and tiedown attachments. Lifting and tiedowns attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the boiler. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the boiler on the carrier when shipped.

3.25 Mounted operating instructions. Unless otherwise specified (see 6.2), operating instructions shall be summarized on a decal or corrosion-resistant plate to be mounted inside the control panel door or similar weatherproofed location.

3.26 Identification marking. Identification shall be permanently and legibly marked directly on the boiler or on a corrosion-resistant metal plate. Identification shall include manufacturer's name, boiler model and serial number, thermal capacity, maximum allowable working pressure, ASME code stamp, year manufactured, and other pertinent information as identified in the ASME codes. The Department of Defense contract number that this equipment is procured under shall also be marked on the equipment or identification plate.

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3.27 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each boiler. When furnished, the spare parts required and the quantity thereof shall be as specified (see 6.2).

3.28 Factory start-up service. When specified (see 6.1 and 6.2), factory authorized service personnel shall be provided to supervise start up of the packaged boiler for five 8-hour working days.

3.29 Workmanship. Workmanship shall be of such quality as to produce boilers in accordance with the ASME Code, as applicable and shall meet the requirements of this specification and standards prevailing among manufacturers who normally produce this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Documentation of inspection shall be made available to the government upon request.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).
b. On-site inspection (see 4.2.2).

4.2.1 Quality conformance inspection. This inspection shall be performed at the manufacturer's facilities. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5 and 4.6, and the packaging inspection of 4.8.

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4.2.2 On-site inspection. The on-site inspection shall be in addition to the quality conformance inspection performed at the manufacturer's facilities. Unless otherwise specified (see 6.2), the on-site inspection shall consist of the tests of 4.7. All or part of the on-site inspection may be performed at the manufacturer's facilities when specified by the contracting officer (see 6.2). When specified (see 6.2), the examination of 4.3 and either the tests of 4.5 or 4.6, or both, shall be repeated on-site.

4.3 Examination. Each boiler and related equipment supplied under this specification shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability.

4.4 Inner casing. Combustion chambers and flue-gas passages of boilers normally operated under pressure shall be tested for tightness at either 1.5 times the design pressure of the weakest section to be tested or 20 inches (508 mm) of water, whichever is less. The test shall be performed prior to the installation of the insulation and outer casing. The section of the unit to be tested shall be carefully sealed using suitable means for blanking any area not to be subjected to test pressure. Air shall be admitted to the unit until the test pressure is reached and then the unit shall be carefully sealed. The outer casing shall be vented to the atmosphere during the casing pressure test. If the pressure does not drop greater than 5 inches (127 mm) of water in ten minutes, the unit shall be regarded as satisfactorily tight and accepted. Failure to meet the requirements of 3.9.8 shall constitute cause for rejection. Any change in air temperature which affects the test conditions shall be taken into consideration.

4.5 Fuel systems. The preassembled fuel system that is furnished integrally with the boiler shall be checked for leakage during the fire testing of the boiler. The soap bubble or the halogen-sniff test method (ASTM E 515 or ASTM E 427 respectively) shall be used to verify integrity of the gas fuel system. Failure to meet the requirements of 3.12 shall constitute cause for rejection.

4.6 Relief valve set pressure test. Relief valves required by the ASME Code shall be individually tested on a test bench using a gas to confirm the required set pressure. The test shall be conducted at a facility that normally performs this work. The relief valve shall be tested on site after the unit is hooked up and ready for operation when specified (see 4.2.2).

4.7 On-site tests. Operational fire tests and auxiliary equipment tests shall be performed on each installed boiler. These tests shall be in addition to any manufacturer's standard fire test procedure, unless compliance evidence provided by the boiler manufacturer is accepted at the discretion of the contracting officer or authorized representative. Acceptable fire test results from identical or similar boiler equipment installed at the same site may be used at the discretion of the contracting officer as proof that equipment procured under this specification is in compliance with one or more of the tests of 4.7.1 through 4.7.9.

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4.7.1 *Operational tests.* The boiler shall be tested to demonstrate control and operational conformance to the requirement of this specification and F-B-2910 or F-S-2911, as applicable, under varying load conditions ranging from the specified maximum capacity to the minimum turndown ratio without on-off cycling. Unless otherwise specified (see 6.2), oil burners shall be tested for ability to make a cold start from the fuel oil supply temperature specified in 3.19. Automatic burners shall also be tested for ability to provide steam in accordance with steam demand when on-off cycling is required. The economizer, when furnished, shall satisfy the requirements of 3.17. Boilers that are set up and operated in accordance with the manufacturer's recommendations and exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected.

4.7.2 Capacity and efficiency tests. The capacity, efficiency, and steam quality at the specified maximum capacity of the boiler shall be determined in accordance with the abbreviated efficiency test of ASME PTC 4.1, using the heat loss method. Performance shall be as specified in 3.7. Methods of measuring the flow of steam, water, and fuel other than weighing will be allowed. The accuracy of the method shall assure the Government the desired performance specified in 3.7 and be acceptable to the contracting officer or his authorized representative. Failure to meet the requirements of 3.7 shall constitute cause for rejection.

4.7.3 Equipment tests. Control panels, preheaters, insulation, valves, controllers, regulators, soot blowers, electric motors, dampers, and other boiler related equipment shall be operated as part of the capacity and efficiency tests. The operation of this equipment shall be closely observed during the operational tests for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth without backlash.

4.7.4 Sequencing. The boiler shall start, operate, and stop in strict compliance with the specified operating sequence.

4.7.5 Burner tests. Tests shall be conducted on the burner as specified in F-B-2910 and as specified herein.

4.7.6 Stoker tests. Tests shall be conducted on the stoker as specified in F-S-2911 and as specified herein.

4.7.7 Boiler limit and cutout switches. Safety shutdowns shall be caused by simulating fuel safety interlock actuating conditions for each boiler limit and cutout switch specified in 3.10. The safety shutdowns shall occur without damage to the equipment.

4.7.8 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.7.9 Emissions testing. The emissions testing for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide specified in 3.8 shall be conducted in accordance with the procedures of the governing federal, state, or local environmental agency, as specified (see 6.2).

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5. PRESERVATION AND PACKING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Selection factors. The following should be considered in the selection of a boiler and the exercise of options herein:

a. The boiler type (oil, gas, or combination) should be determined in accordance with the availability or economics of the fuel supply and any DoD requirements for capability of firing multiple fuels (see 3.1).
b. The usual commercial lower limit of this style of boiler is about 6000 PPH (2722 kgph).
c. Normally, oil fired boilers above 6000 PPH (2722 kgph) output should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6. When changing grade of fuel, replacement may be necessary (1) of the fuel nozzles because of the different fuel viscosity, and (2) of the fuel pump motor to drive the pump at a speed that ensures required fuel pressure in order to maintain the required burner output and meet the optimum air to fuel ratio (see 3.1).
d. Stoker fired only boilers shall have a thermal output capacity range between 2000 and 50 000 PPH (907 and 22 680 kgph).
e. Retrofitting a gas or oil fired boiler with a stoker will require the front wall to be removed and replaced with a stoker front. A cinder return system may be added if economical. Refractory may be added in back of the bottom portion of the packaged boiler. This provision will add a significant cost to the packaged boiler. Local environmental regulations and coal availability should be considered before specifying this design requirement (see 3.6.1).
f. If the boiler is subject to environmental regulations, each agency should consult with their environmental support office since regulation limits of emissions vary between federal, state, and local agencies (see 3.8).
g. Oxygen compensation systems should not be specified for fixed rate on-off, or three position high-low combustion controls (see 3.15.1).
h. A packaged boiler mounted preheater should be specified only at installations of single heavy oil fired boilers. It is more economical for multiple boiler installations to use a common preheater that is external from the individual packaged boiler skids (see 3.19).
i. Electric preheaters should be furnished on units firing No. 4 or No. 5 (light) fuel oil if required for proper atomization in low pressure, air atomizing systems (see 3.19).
j. Factory service should be specified only if the boiler will be installed and tested without a long period of interim storage (see 3.28).

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k. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the boiler is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

6.1.1 Coal characteristics. Lignite or sub-bituminous coal is characterized by lower Btu content by weight and agglomerating tendencies. Nonagglomerating coal with a Btu (cal-kg) content less than 11 500 Btu per pound (7 764 cal-kg/kg) fall in this category.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, date, and number of this specification.
b. Types, sizes, and classes of boilers required (see 1.2).
c. Required steam output (see 3.1).
d. Fuel required (see 3.1).
e. Whether the boiler is to be furnished with a burner as specified in F-B-2910, or a stoker as specified in F-S-2911, or both (see 3.1).
f. Design parameters for boilers (see 3.6):
g. When the boiler is to be designed for retrofit with a stoker (see 3.6.1).
h. Volume heat input rates for coal fired boilers, if other than as specified (see 3.7.3.2).
i. When an economizer is required (see 3.7.4).
j. When the manufacturer is to provide the following information for design and installation of a flue gas baghouse (see 3.7.5):
k. Emission requirements for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide (see 3.8).
l. Clearances and construction of radiant tubes, if other than as specified (see 3.9.2).
m. Length of furnace, if other than as specified (see 3.9.6).
n. Trim and appurteriances to be supplied with boiler, if other than as specified (see 3.10).
o. When a manual feedwater regulator control is to be furnished (see 3.11.2).
p. When the combustion control system is to be furnished with an oxygen compensation system, or an oxygen compensation system with an unburned combustible gas analyzer; and type of burner combustion control system required (see 3.15.1).
q. Location of control panel (see 3.16).
r. When an economizer is required (see 3.17).
s. When automatically controlled sequencing soot blowers are to be provided (see 3.18).
t. When a fuel preheater is required; if required, type of preheater to be provided (see 3.19).
u. Fuel oil supply temperature (see 3.19).
v. When the steam preheater is to be furnished with an electric preheater (see 3.19.2.1).
w. When a cinder return system is required for class 6, class 7, class 8, or class 9 boilers (see 3.20).
x. When the standard transition piece is to be furnished (see 3.21).
y. Required voltage, phase, and frequency for operation of motors; and when motors of 100 horsepower and less are not to be provided with across-the-line starter and overload protection (see 3.22).

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z. When lifting and tiedown attachments are required (see 3.24).
aa. Operating instructions, if other than as specified (see 3.25).
bb. When spare parts are to be required; if required, the type and the quantity to be provided (see 3.27).
cc. When factory start up service is required (see 3.28).
dd. On-site inspection, if other than as specified; when all or part of the on-site inspection may be performed at the manufacturer's facility; and whether the examination of 4.3 and with the tests of 4.5 or 4.6, or both, are to be repeated after boiler delivery on-site (see 4.2.2).
ee. When oil burners are not to be tested for ability to make a cold start (see 4.7.1).
ff. Testing procedures of the governing federal, state, or local environmental agency for opacity, particulates, nitrogen oxides, carbon monoxide, and sulfur dioxide (see 4.7.9).
gg. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 227.405-70 are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 Standards compliance. Prior to approval of shipment, the contractor shall submit to the contracting officer or authorized representative satisfactory evidence that the proposed boiler to be furnished under this specification meets the requirements of the ASME Code, NFPA 70, NFPA 85C, and UL-353.

6.4.1 ASME Code requirements. Acceptable evidence of meeting the requirements of Section I, Section IV, Section IX, and when an economizer or preheater is required, Section VIII, Division 1 of the ASME Code is the ASME Code symbol.

6.4.2 NFPA requirements. Acceptable evidence of meeting the requirements of NFPA 70 and NFPA 85C is the receipt of the manufacturer's certificate of compliance stating the equipment procured with this specification is in accordance with the applicable requirements of NFPA 70, NFPA 85C.

6.4.3 UL requirements. Acceptable evidence of meeting the requirements of UL-353 is the certification symbol, or listing mark on the products, or a certified test report from a recognized independent testing laboratory indicating the boiler controls have been tested and conform to UL requirements.

6.4.4 NEMA requirements. Acceptable evidence of meeting the requirements of NEMA ICS 1, NEMA ICS 2, NEMA ICS 6, and NEMA MG-1, is the receipt of the manufacturer's certificate of compliance stating the equipment procured under this specification is in accordance with the applicable requirements of NEMA ICS 1, NEMA ICS 2, NEMA ICS 6, and NEMA MG-1.

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6.5 Stoker retrofitting instructions. The manufacturer shall submit instructions for retrofitting the boiler with the recommended stoker for converting to coal firing (see 3.6.1).

6.6 Design verification. The manufacturer shall furnish data to verify compliance with the specified heat input rates (see 3.7.3).

6.7 Baghouse installation. Installation instructions covering the connection of the flue gas baghouse shall be provided (see 3.7.5).

6.8 Remote control panel installation. When a free-standing remote mounted control panel is to be furnished (see 3.16), installation instructions covering the connection of the control panel shall be provided.

6.9 Relief valve. A certificate of compliance verifying the performance and results of the relief valve pressure test (see 4.6) shall be provided to the contracting officer.

6.10 Technical manuals. The manufacturer shall provide two commercial technical manuals normally prepared and supplied with the boiler.

6.11 Definitions. Definitions used in this specification are in accordance with the American Boiler Manufacturers Association Lexicon of Boiler and Auxiliary Equipment. Other terms used in this specification should be interpreted in accordance with the referenced national standards.

6.12 Boiler operational certification (Navy only). The boiler operational certification should be performed, where applicable, per NAVFAC MO-324 after the boiler complies with the on-site tests (see 4.7). The boiler inspector should be certified by NAVFAC and licensed by the Engineering Field Division.

6.13 Subject term (key word) listing.

Burner
Coal fired
Combination fuel fired
Gas fired
Oil fired
Stoker

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MILITARY INTERESTS:

Custodians:
Navy - YD1
Army - CE
Air Force - 99

Review Activities:
Navy - MC, SA
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:

Navy - YD1

(Project 4410-0098)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER F-B-2902	2. DOCUMENT DATE (YYMMDD) 960722

3. DOCUMENT TITLE
BOILERS, STEAM WATERTUBE (BENT TUBE, MULTI-DRUM AND CROSS DRUM) PACKAGED TYPE (10,000,000 TO 125,000,000 BTU/HR THERMAL OUTPUT CAPACITY)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
RUSSELL REYNOLDS	(1) Commercial: (805) 982-5946 (2) AUTOVON: 551-5946
c. ADDRESS (Include Zip Code) COMMANDING OFFICER, NCBC CODE 1581C 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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